A MAP FROM A COMPACT OBJECT INTO A NICELY FILTERED DG MODULE FACTORS THROUGH A FINITELY FILTERED SUBMODULE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-factor-through-nicer (source lines 6433-6451); the proof environment begins two lines below it and is its own proof, closing at source line 6495.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the compactness and filtration notions are part of the author's surrounding development in the same chapter and are named rather than cited by label.

Editorial note. The proof is an induction whose hypothesis is stated inside the proof itself; the three listed clauses of that hypothesis are the author's own wording and are reproduced verbatim.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: a map from a compact object into a nicely filtered dg module factors through a submodule with finite direct-sum quotients.

Allowed standard prerequisites (editorial).

- Compact object of a triangulated category. An object E is compact when the functor it represents commutes with arbitrary direct sums, so that any morphism from E into a direct sum factors through a finite subsum.. It is invoked at source lines 6458-6462 to enlarge the finite index set so that E -> P factors through the constructed submodule.
- Finite filtration of a dg module by shifted copies of A. The module P carries a finite increasing filtration by dg submodules whose successive quotients are direct sums of A shifted by integers.. The induction runs down the filtration levels and the quotient description is what is preserved at every step (source lines 6436-6440, 6465-6470).
- Homogeneous generators and the differential on a quotient piece. Each graded piece is free on homogeneous generators of known degrees, and the differential of a generator decomposes into a combination of the other generators plus a term in the next lower filtration level.. The choice of the x_{i,j} and the displayed formula for d(x_{i,j}) drive the construction (source lines 6456-6465).

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Lemma 0.1. *Let (A, d) be a differential graded algebra. Let E be a compact object of $D(A, d)$. Let P be a differential graded A-module which has a finite filtration*

$$0 = F_{-1}P \subset F_0P \subset F_1P \subset \ldots \subset F_nP = P$$

by differential graded submodules such that

$$F_{i+1}P/F_iP \cong \bigoplus\nolimits_{j \in J_i} A[k_{i,j}]$$

as differential graded A-modules for some sets J_i and integers $k_{i,j}$. Let $E \to P$ be a morphism of $D(A, d)$. Then there exists a differential graded submodule $P' \subset P$ such that $F_{i+1}P \cap P'/(F_iP \cap P')$ is equal to $\bigoplus_{j \in J'_i} A[k_{i,j}]$ for some finite subsets $J'_i \subset J_i$ and such that $E \to P$ factors through P'.

Proof. We will prove by induction on $-1 \leq m \leq n$ that there exists a differential graded submodule $P' \subset P$ such that

(1) $F_mP \subset P'$,
(2) for $i \geq m$ the quotient $F_{i+1}P \cap P'/(F_iP \cap P')$ is isomorphic to $\bigoplus_{j \in J'_i} A[k_{i,j}]$ for some finite subsets $J'_i \subset J_i$, and
(3) $E \to P$ factors through P'.

The base case is $m = n$ where we can take $P' = P$.

Induction step. Assume P' works for m. For $i \geq m$ and $j \in J'_i$ let $x_{i,j} \in F_{i+1}P \cap P'$ be a homogeneous element of degree $k_{i,j}$ whose image in $F_{i+1}P \cap P'/(F_iP \cap P')$ is the generator in the summand corresponding to $j \in J_i$. The $x_{i,j}$ generate P'/F_mP as an A-module. Write

$$\mathrm{d}(x_{i,j}) = \sum x_{i',j'} a_{i,j}^{i',j'} + y_{i,j}$$

with $y_{i,j} \in F_mP$ and $a_{i,j}^{i',j'} \in A$. There exists a finite subset $J'_{m-1} \subset J_{m-1}$ such that each $y_{i,j}$ maps to an element of the submodule $\bigoplus_{j \in J'_{m-1}} A[k_{m-1,j}]$ of $F_mP/F_{m-1}P$. Let $P'' \subset F_mP$ be the inverse image of $\bigoplus_{j \in J'_{m-1}} A[k_{m-1,j}]$ under the map $F_mP \to F_mP/F_{m-1}P$. Then we see that the A-submodule

$$P'' + \sum x_{i,j} A$$

is a differential graded submodule of the type we are looking for. Moreover

$$P'/(P'' + \sum x_{i,j}A) = \bigoplus\nolimits_{j \in J_{m-1} \setminus J'_{m-1}} A[k_{m-1,j}]$$

Since E is compact, the composition of the given map $E \to P'$ with the quotient map, factors through a finite direct subsum of the module displayed above. Hence after enlarging J'_{m-1} we may assume $E \to P'$ factors through $P'' + \sum x_{i,j}A$ as desired. $\square$